

Question number	Answer	Mark
8(a)	so other scientists could repeat the study / because different species might respond differently to microgravity	(1)

Question number	Answer	Mark
8(b)	nematodes may be {harmed / killed}	(1)

Question number	Answer	Mark
8(c)	A description that includes the following points: - epigenetic changes to {prevent / allow} gene expression (1) - transcription factors to switch {on / off} genes (1) - post-transcriptional modification to determine which protein is produced from a particular gene (1)	(3)

Question number	Answer	Mark
8(d)	An answer that includes any three of the following points: • (lower fat accumulation because) the diet that the nematodes were fed did not have an energy content greater than that being used (1) • (lower fat accumulation because) nematodes did not need to store energy as they were not very active (1) • (shorter body length because) diet was low in protein (1) • genes controlling {fat accumulation / body length} had been switched off (1) • microgravity resulted in less protein being synthesised (1)	(3)

Question number	Answer	Mark
8(e)	An explanation that includes any four of the following points: • less muscle {protein / named protein} synthesised because of muscle atrophy (1) • because the microgravity decreases the force that needs to be exerted by the muscles (1) • less muscle activity, therefore less ATP required (1) • therefore fewer {enzymes / named enzyme} needed for respiration (1) • fewer proteins involved in gluconeogenesis because less glucose needed for respiration (1)	(4)

Question number	Answer	Mark
8(f)	An explanation that includes the following points: • microarrays analyse the {active DNA / RNA} of a cell (1) • therefore less {active DNA / RNA} for the electron transport genes and increased {active DNA / RNA} for the sirtuin gene (1) • compared to nematodes, they are not exposed to microgravity (1)	(3)

Question number	Answer	Mark
8(g)	An explanation that includes the following points: • because the scientists wanted to assess the lipid stores of the nematodes (1) • Sudan Black binds specifically to fat (1)	(2)

Question number	Answer	Mark
8(h)	An answer that includes three of the following points: • mean values are given throughout (1) • frequency data reliable as the standard deviations do not overlap (1) • frequency and wavelength data reliable as p values are less than {0.01 / 0.05} (1) • they are not very reliable as the sample sizes are very small (1)	(3)